

# FPGA experiment report

Task: FPGA 3

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## Part 1 – Status update

Fill in the following table according to your progress at the time of the submission.

Each row should refer to one of the modules together with its simulation.

The status column should be filled with one of the following:

- Done – If both are written and simulation passed.
- Partially done – If you started writing the module and simulation but it's not working properly.
- Not started – If you didn't get to that part of the task.

When the status is *partially done*, please use the notes/comments column to explain in a few more words what is the status of the module and what is the status of the simulation.

Fill here:

| Module name/Step        | status | notes/comments |
|-------------------------|--------|----------------|
| Seven Segment Display   | Done   |                |
| Display Playground      | Done   |                |
| Setting up constraints  | Done   |                |
| Implementing the design | Done   |                |

### Issues:

In this section please describe the current issues you are facing.

Per each module and simulation, if you receive any errors at the time of submission, please describe which errors and what they mean.

Fill here:

Seven Segment Display: None

Display Playground: None

Setting up constraints: None

Implementing the design: None

## **Part 2 – Simulations and answers**

In this section, please add the images for each simulation as described in the report. Remember to use annotations and explain in your own words what is shown in each simulation.

In some of the tasks in the report, there are also questions that you should answer here.

Make sure to include images of all simulations running at the time of submission, even if they are not correct.

Fill here:

### **Seven Segment Display:**

#### **2c. answer:**

If digit advance occurs every  $2^k$  clocks:

$$T_{\text{digit}} = \frac{2^k}{f_{\text{clk}}}, \quad T_{\text{display}} = \frac{4 \cdot 2^k}{f_{\text{clk}}}$$

With  $s = \text{clkdiv}[19:18]$ , the LSB of  $s$  is bit 18, so  $k = 18$  and  $f_{\text{clk}} = 100 \text{ MHz} = 10^8 \text{ Hz}$ .

#### **Time per digit:**

$$T_{\text{digit}} = \frac{2^{18}}{10^8} = \frac{262,144}{10^8} = 2.62 \text{ ms}$$

#### **Full display time:**

$$T_{\text{display}} = \frac{4 \cdot 2^{18}}{10^8} = 4 \times 2.62 \text{ ms} = 10.49 \text{ ms}$$

$$1 \text{ ms} \leq 10.49 \text{ ms} \leq 16 \text{ ms} \quad \checkmark$$

Therefore, using bits [19:18] is a good choice.

## Display Playground:

### 3b. answer:

The Display Playground module consists of the following functional blocks:

- **Input clamping logic:** Four parallel comparators check whether each 4-bit switch group ( $sw[3:0]$ ,  $sw[7:4]$ ,  $sw[11:8]$ ,  $sw[15:12]$ ) exceeds the value 9. If so, a multiplexer replaces the value with 9 ( $4'b1001$ ). Otherwise the original switch value passes through.
- **Seg\_7\_Display instance:** The clamped 16-bit value `fixed_sw` is fed to the `Seg_7_Display` module, which handles the time-multiplexed driving of the four 7-segment digits.
- **LED decoder:** A case statement maps the first digit (`fixed_sw[3:0]`) to the corresponding LED (0–9), producing a hot representation on the 10 LEDs.

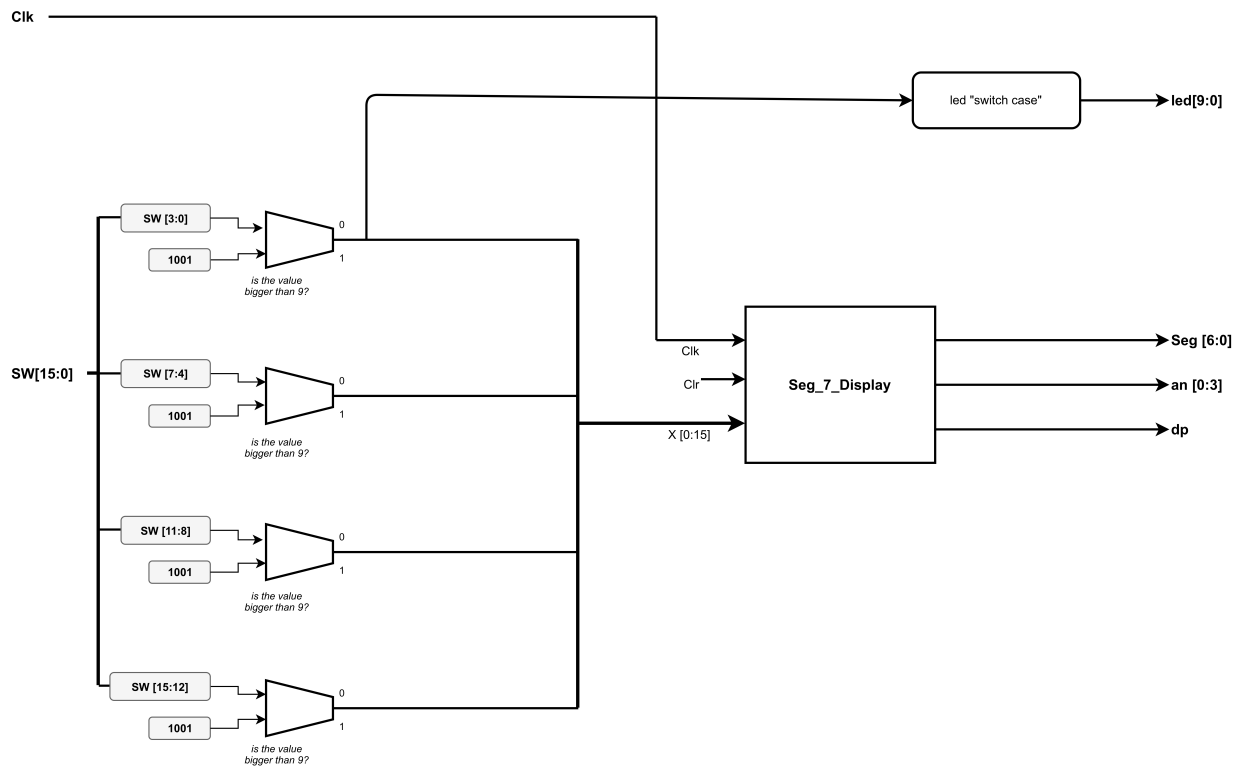


Figure 1: Display Playground block diagram

### 3c. answer:

If the switches indicate a number greater than 9 (for example,  $4'b1101$  which equals 13 in decimal), the design clamps the value to 9, the maximum single decimal digit. This is implemented using a generate loop with comparators: for each 4-bit group, if the value exceeds  $4'd9$ , it is replaced by  $4'd9$ . This prevents undefined behavior on the 7-segment display and ensures only digits 0–9 are shown.

### 3f. answer:

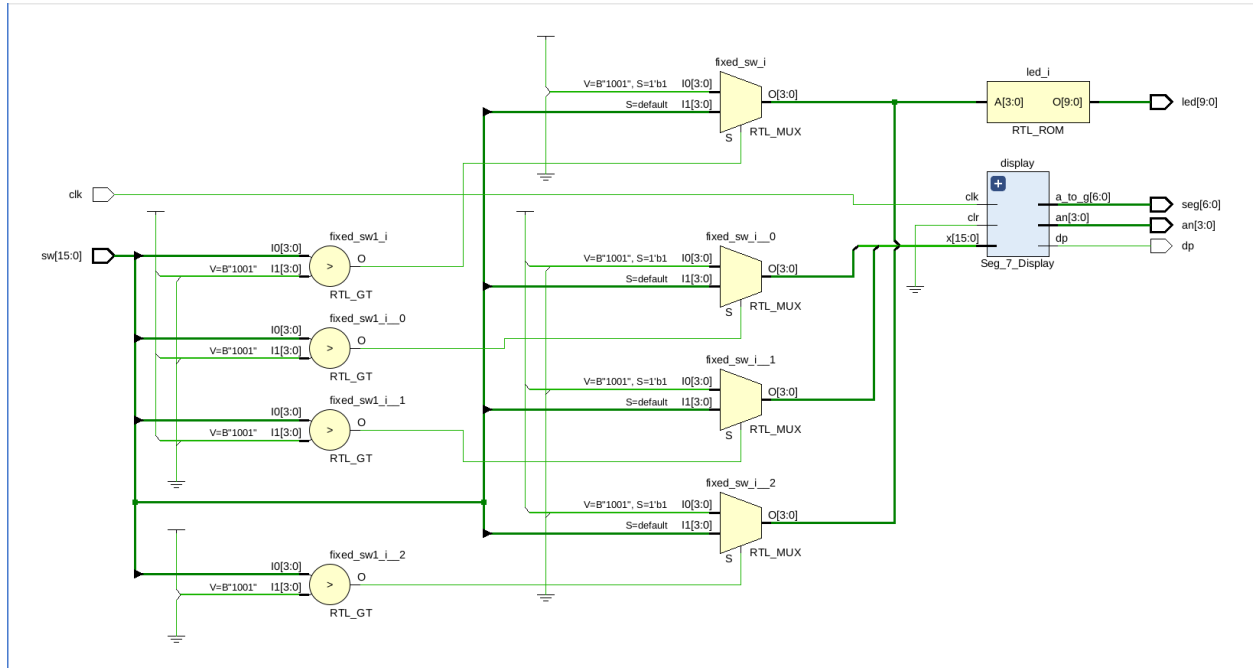


Figure 2: RTL elaborated design schematic of Display\_Playground. The schematic shows 2 inputs clk, [15:0] sw and four outputs [6:0] seg, [3:0] an, dp, [9:0] led four RTL\_GT, comparators checking each 4-bit switch group against 9, four RTL\_MUX multiplexers selecting between the original and clamped values, an RTL\_ROM for the LED decoder, and the instantiated Seg\_7\_Display module.

3g. answer:

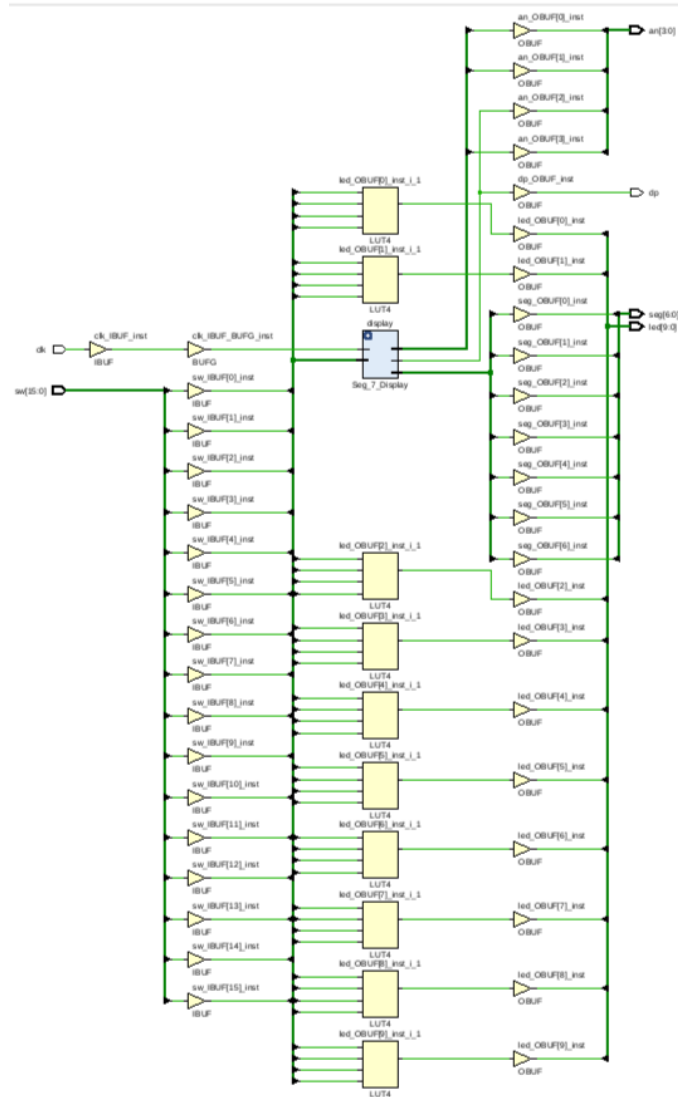


Figure 3: Synthesized design schematic of Display\_Playground. The comparators and muxes have been optimized into LUT4 look-up tables. Input buffers (IBUF) and output buffers (OBUF) have been inserted for the physical I/O. The clock input passes through a BUFG global clock buffer.

In the elaborated design we see the logical structure of the design as written in Verilog, before any optimization. We can see how modules, registers and logic blocks are connected based on the Verilog code we wrote.

In the synthesized design schematic we see the actual hardware resources that the FPGA will use to implement the Verilog code. The high-level constructs (comparators, multiplexers, ROM) are mapped to FPGA primitives such as LUT4 look-up tables, and I/O buffer primitives (IBUF, OBUF, BUFG) are inserted for all external connections.

## Setting up constraints:

The constraints file `fpga_lab_constraints.xdc` was configured as follows:

1. **Clock signal:** Uncommented the two lines connecting `clk` to pin `W5` with LVCMOS33 I/O standard.
2. **Timing constraint:** Uncommented the `create_clock` command specifying a 10 ns period (100 MHz) for `sys_clk_pin`.
3. **Switch inputs:** Uncommented all 16 switch lines (`sw[0]` through `sw[15]`).
4. **7-segment outputs:** Uncommented the 7 segment lines (`seg[0]`–`seg[6]`), the 4 anode lines (`an[0]`–`an[3]`), and the decimal point (`dp`).
5. **LED outputs:** Uncommented 10 LED output pins (`led[0]`–`led[9]`) mapped to pins `U16`, `E19`, `U19`, `V19`, `W18`, `U15`, `U14`, `V14`, `V13`, and `V3`.

## Implementing the design:

5c. answer:

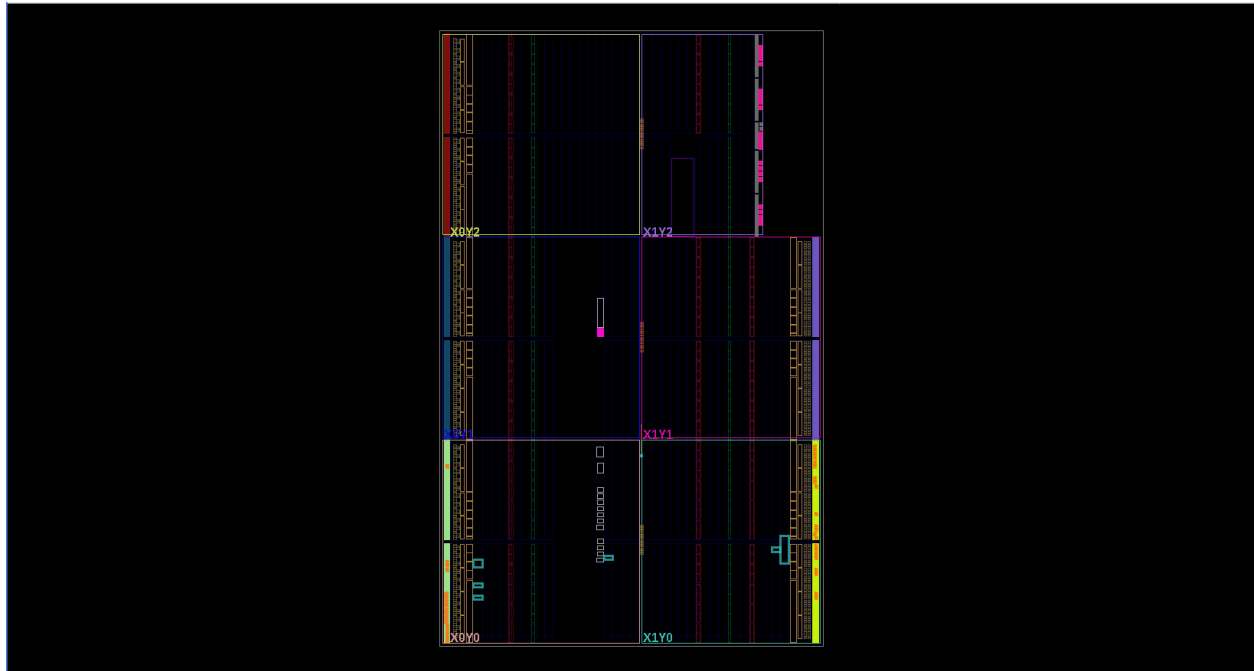


Figure 4: FPGA implementation floorplan view showing the placed and routed design on the Artix-7 device. The utilized resources are concentrated in a small region of the chip.

## 5d. answer:

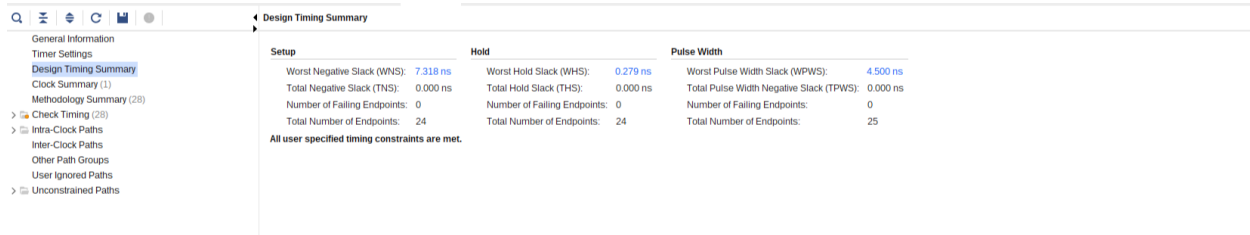


Figure 5: Design Timing Summary showing positive slack for both setup and hold times. All user-specified timing constraints are met.

The timing summary confirms:

- **Setup:** Worst Negative Slack (WNS) = 7.318 ns (positive  $\Rightarrow$  constraint met)
- **Hold:** Worst Hold Slack (WHS) = 0.279 ns (positive  $\Rightarrow$  constraint met)
- **Pulse Width:** Worst Pulse Width Slack (WPWS) = 4.500 ns (positive  $\Rightarrow$  constraint met)
- Number of Failing Endpoints: 0 for all categories

## 5e. answer:

The critical setup path length can be calculated from the worst negative slack:

$$\text{setup slack} = (T_{clk}) - (T_{critical\ path})$$

$$7.318 = 10 - (T_{critical\ path})$$

$$T_{critical\ path} = 2.682 \text{ [ns]}$$

Another way to find the critical path is from the Vivado software:

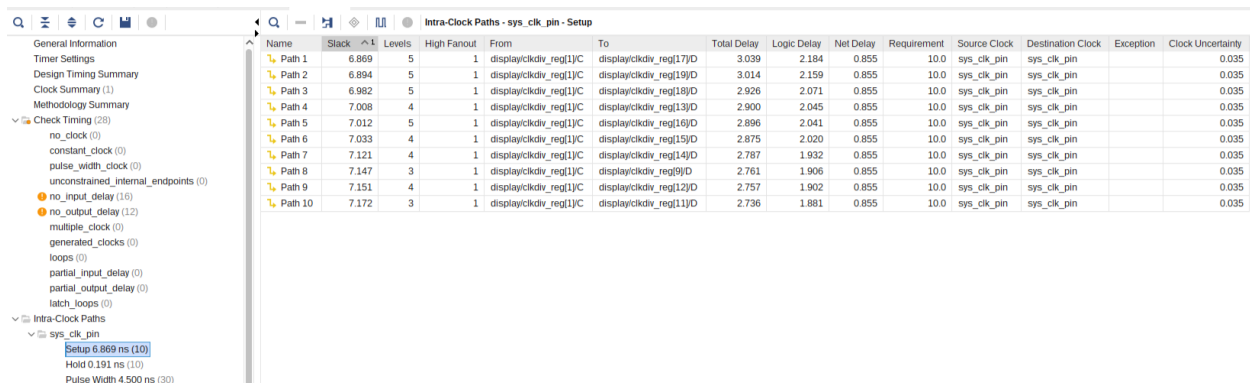


Figure 6: Intra-Clock Paths showing the critical path from display/clkdiv\_reg[1]/C to display/clkdiv\_reg[17]/D with a total delay of 3.039 ns.

We can see that the critical path is 3.039 [ns] which is close to the value we calculated. The small difference is due to the inclusion of clock uncertainty (0.035 ns) in the slack computation.

## FPGA programming

### 6a–b. Bitstream Generation and Programming:

The bitstream was generated successfully and the design was programmed onto the BASYS3 board.

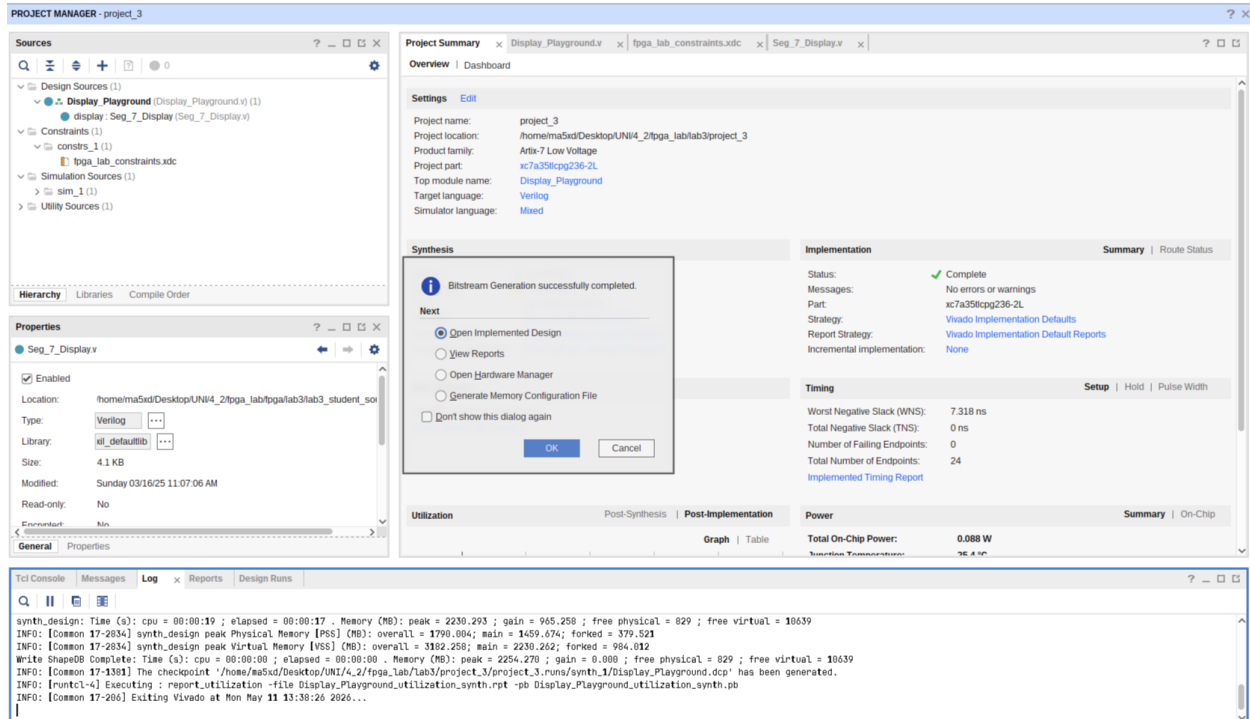


Figure 7: Project Summary showing successful bitstream generation. Implementation status is “Complete” with no errors or warnings. WNS = 7.318 ns, Total On-Chip Power = 0.088 W.

The design was verified on hardware: setting the 16 switches to represent four decimal digits correctly displayed them on the 7-segment display, and the corresponding LED for the first digit lit up.



## 6c. answer:

After removing the debug core and re-implementing the design:

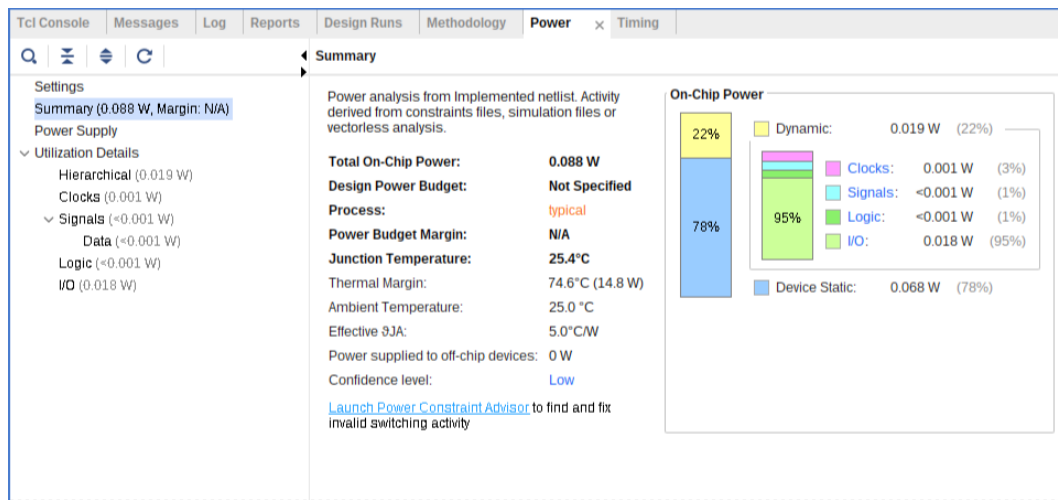


Figure 8: Power report summary. Total On-Chip Power = 0.088 W, consisting of 0.019 W dynamic (22%) and 0.068 W static (78%). Junction temperature = 25.4°C.

| Report                                                    | Type                     | Options                                                                | Modified         | Size     |
|-----------------------------------------------------------|--------------------------|------------------------------------------------------------------------|------------------|----------|
| Synthesis                                                 |                          |                                                                        |                  |          |
| Synth Design (synth_design)                               |                          |                                                                        |                  |          |
| Utilization - Synth Design                                | report_utilization       |                                                                        | 5/11/26, 1:44 PM | 8.1 KB   |
| synthesis_report                                          |                          |                                                                        | 5/11/26, 1:44 PM | 18.6 KB  |
| Implementation                                            |                          |                                                                        |                  |          |
| impl_1                                                    |                          |                                                                        |                  |          |
| Design Initialization (init_design)                       |                          |                                                                        |                  |          |
| Timing Summary - Design Initialization                    | report_timing_summary    | max_paths = 10; report_unconstrained = true;                           |                  |          |
| Opt Design (opt_design)                                   |                          |                                                                        |                  |          |
| DRC - Opt Design                                          | report_drc               |                                                                        | 5/11/26, 1:47 PM | 1.4 KB   |
| Timing Summary - Opt Design                               | report_timing_summary    | max_paths = 10; report_unconstrained = true;                           |                  |          |
| Power Opt Design (power_opt_design)                       |                          |                                                                        |                  |          |
| Timing Summary - Power Opt Design                         | report_timing_summary    | max_paths = 10; report_unconstrained = true;                           |                  |          |
| Place Design (place_design)                               |                          |                                                                        |                  |          |
| IO - Place Design                                         | report_io                |                                                                        | 5/11/26, 1:47 PM | 72.3 KB  |
| Utilization - Place Design                                | report_utilization       |                                                                        | 5/11/26, 1:47 PM | 9.9 KB   |
| Control Sets - Place Design                               | report_control_sets      | verbose = true;                                                        | 5/11/26, 1:47 PM | 3.8 KB   |
| Incremental Reuse - Place Design                          | report_incremental_reuse |                                                                        |                  |          |
| Incremental Reuse - Place Design                          | report_incremental_reuse |                                                                        |                  |          |
| Timing Summary - Place Design                             | report_timing_summary    | max_paths = 10; report_unconstrained = true;                           |                  |          |
| Post-Place Power Opt Design (post_place_power_opt_design) |                          |                                                                        |                  |          |
| Timing Summary - Post-Place Power Opt Design              | report_timing_summary    | max_paths = 10; report_unconstrained = true;                           |                  |          |
| Post-Place Phys Opt Design (phys_opt_design)              |                          |                                                                        |                  |          |
| Timing Summary - Post-Place Phys Opt Design               | report_timing_summary    | max_paths = 10; report_unconstrained = true;                           |                  |          |
| Route Design (route_design)                               |                          |                                                                        |                  |          |
| DRC - Route Design                                        | report_drc               |                                                                        | 5/11/26, 1:48 PM | 1.4 KB   |
| Methodology - Route Design                                | report_methodology       |                                                                        | 5/11/26, 1:48 PM | 6.7 KB   |
| Power - Route Design                                      | report_power             |                                                                        | 5/11/26, 1:48 PM | 8.3 KB   |
| Route Status - Route Design                               | report_route_status      |                                                                        | 5/11/26, 1:48 PM | 0.6 KB   |
| Timing Summary - Route Design                             | report_timing_summary    | max_paths = 10; report_unconstrained = true;                           | 5/11/26, 1:48 PM | 203.1 KB |
| Incremental Reuse - Route Design                          | report_incremental_reuse |                                                                        |                  |          |
| Clock Utilization - Route Design                          | report_clock_utilization |                                                                        | 5/11/26, 1:48 PM | 11.1 KB  |
| Bus Skew - Route Design                                   | report_bus_skew          | warn_on_violation = true;                                              | 5/11/26, 1:48 PM | 1.0 KB   |
| Implementation Log                                        |                          |                                                                        | 5/11/26, 1:48 PM | 43.6 KB  |
| Post-Route Phys Opt Design (post_route_phys_opt_design)   |                          |                                                                        |                  |          |
| Timing Summary - Post-Route Phys Opt Design               | report_timing_summary    | max_paths = 10; report_unconstrained = true; warn_on_violation = true; |                  |          |
| Bus Skew - Post-Route Phys Opt Design                     | report_bus_skew          | warn_on_violation = true;                                              |                  |          |
| Write Bitstream (write_bitstream)                         |                          |                                                                        |                  |          |
| Implementation Log                                        |                          |                                                                        | 5/11/26, 1:48 PM | 43.6 KB  |

Figure 9: Vivado Reports tab showing all synthesis and implementation reports generated successfully.

## Appendix – Verilog Source Code

### Seg\_7\_Display.v

```
1  `timescale 1ns/10ps
2  ///////////////////////////////////////////////////////////////////
3  // Company:           Tel Aviv University
4  // Module Name:       Seg_7_Display
5  // Description:       Translates input vector "x" into 7-segment
6  //                    display signals with time-multiplexed digits
7  ///////////////////////////////////////////////////////////////////
8  module Seg_7_Display(
9      input [15:0] x,
10     input clk,
11     input clr,
12     output reg [6:0] a_to_g,
13     output reg [3:0] an,
14     output wire dp
15 );
16
17     wire [1:0] s;
18     reg [3:0] digit;
19
20     // For 100MHz clock
21     reg [19:0] clkdiv = 19'b0;
22     assign s = clkdiv[19:18];
23
24     assign dp = (s == 2'b10) ? 0 : 1;
25
26     always @(posedge clk)
27         case(s)
28             0: digit = x[3:0];
29             1: digit = x[7:4];
30             2: digit = x[11:8];
31             3: digit = x[15:12];
32             default: digit = x[3:0];
33         endcase
34
35     // 7-segment decoder
36     always @(*)
37         case(digit)
38             0: a_to_g = 7'b1000000;
39             1: a_to_g = 7'b1111001;
40             2: a_to_g = 7'b0100100;
41             3: a_to_g = 7'b0110000;
42             4: a_to_g = 7'b0011001;
43             5: a_to_g = 7'b0010010;
44             6: a_to_g = 7'b0000010;
45             7: a_to_g = 7'b1111000;
46             8: a_to_g = 7'b0000000;
47             9: a_to_g = 7'b0010000;
48             'hA: a_to_g = 7'b0111111; // dash
49             'hB: a_to_g = 7'b1111111; // off
50             'hC: a_to_g = 7'b1110111;
51             default: a_to_g = 7'b0000000;
52         endcase
53
54     // Anode selection (active low)
```

```
55     always @(*) begin
56         an = 4'b1111;
57         an[s] = 0;
58     end
59
60     // Clock divider counter
61     always @(posedge clk or posedge clr) begin
62         if (clr == 1)
63             clkdiv <= 0;
64         else
65             clkdiv <= clkdiv + 1;
66     end
67 endmodule
```

## Display\_Playground.v

```
1  `timescale 1ns/10ps
2  ///////////////////////////////////////////////////////////////////
3  // Company:           Tel Aviv University
4  // Module Name:       Display_Playground
5  // Description:       Top module - switches select display values,
6  //                   LEDs indicate first digit
7  // Dependencies:      Seg_7_Display
8  ///////////////////////////////////////////////////////////////////
9  module Display_Playground(clk, sw, seg, an, dp, led);
10
11      input clk;
12      input [15:0] sw;
13      output wire [6:0] seg;
14      output wire [3:0] an;
15      output wire      dp;
16      output reg [9:0] led;
17
18      reg [15:0] fixed_sw;
19
20      genvar i;
21      for (i = 0; i < 4; i = i + 1) begin
22          always @(*) begin
23              if (sw[(i*4)+3:i*4] > 4'd9)
24                  fixed_sw[(i*4)+3:i*4] = 4'd9;
25              else
26                  fixed_sw[(i*4)+3:i*4] = sw[(i*4)+3:i*4];
27          end
28      end
29
30      always @(*) begin
31          led = 0;
32          case(fixed_sw[3:0])
33              4'b0000: led[0] = 1'b1;
34              4'b0001: led[1] = 1'b1;
35              4'b0010: led[2] = 1'b1;
36              4'b0011: led[3] = 1'b1;
37              4'b0100: led[4] = 1'b1;
38              4'b0101: led[5] = 1'b1;
39              4'b0110: led[6] = 1'b1;
40              4'b0111: led[7] = 1'b1;
41              4'b1000: led[8] = 1'b1;
42              4'b1001: led[9] = 1'b1;
43              default: led = 0;
44          endcase
45      end
46
47      Seg_7_Display display(fixed_sw, clk, 0, seg, an, dp);
48
49  endmodule
```